

NORMAL INCIDENCE — Γ AND τ

Reflection coeff. *boundary at z = 0, wave from medium 1*

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \text{ (unitless)}$$

Transmission coeff. *always*

$$\tau = 1 + \Gamma = \frac{2\eta_2}{\eta_2 + \eta_1}$$

General η *always*

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_r \mu_0}{\epsilon_r \epsilon_0}} = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \text{ (}\Omega\text{)}$$

Nonmag. shortcut $\mu_r = 1$ (air, most dielectrics)

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \Rightarrow \eta_1 = \eta_0 \text{ (air, } \epsilon_r = 1\text{)}, \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_{r,2}}}$$

η_0 from μ_0, ϵ_0 $\frac{1}{\epsilon_0} \approx 36\pi \times 10^9$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = \sqrt{(4\pi \times 10^{-7})(36\pi \times 10^9)} = \sqrt{144\pi^2 \cdot 10^2} = 12\pi \text{ }\Omega$$

Low-loss η_2 (poor conductor) $\sigma/(\omega\epsilon) \ll 1$

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon'}} \left(1 + j \frac{\sigma}{2\omega\epsilon'}\right) \text{ (}\Omega\text{)}$$

For boundary Γ/τ , take just $|\eta_2| \approx \eta_0/\sqrt{\epsilon_{r2}}$; the j term is tiny.

Ê PHASOR TEMPLATES

Wave in $+\hat{z}$, incident has unit-vector pol. $\hat{e}$, amplitude a_0 :

$$\mathbf{\hat{E}}^i = a_0 \hat{e} e^{-jk_1 z}$$
$$\mathbf{\hat{E}}^r = \Gamma a_0 \hat{e} e^{+jk_1 z} \text{ (travels in } -\hat{z}\text{)}$$
$$\mathbf{\hat{E}}^t = \tau a_0 \hat{e} e^{-jk_2 z} \text{ (lossless medium 2)}$$
$$\mathbf{\hat{E}}^t = \tau a_0 \hat{e} e^{-\alpha_2 z} e^{-j\beta_2 z} \text{ (lossy medium 2)}$$

Total in medium 1: $\mathbf{\hat{E}}_1 = \mathbf{\hat{E}}^i + \mathbf{\hat{E}}^r$.

$\hat{e}$ by polarization *plug into templates above*

Pol.	$\hat{e}$	Note
Lin. $\hat{x}$	$\hat{x}$	$E_y = 0$
Lin. $\hat{y}$	$\hat{y}$	$E_x = 0$
Lin. at 45°	$(\hat{x} + \hat{y})/\sqrt{2}$	in phase
RHC	$\hat{x} - j\hat{y}$	$\delta = -\pi/2$
LHC	$\hat{x} + j\hat{y}$	$\delta = +\pi/2$
General	$a_x \hat{x} + a_y e^{j\delta} \hat{y}$	elliptical

Γ, τ act on $\hat{e}$ unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted *η -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check: $\%P_r + \%P_t = 100$.
If medium 2 is denser ($\eta_2 < \eta_1$), $|\tau| > 1$ but $\%P_t < 100\%$ because of the η_1/η_2 factor.

LOW-LOSS QUICK PARAMS (med 2)

Check first $\sigma/(\omega\epsilon)$

$$\alpha_2 \approx \frac{\sigma_2}{2} \sqrt{\frac{\mu_2}{\epsilon_2}} = \frac{\sigma_2 \eta_0}{2\sqrt{\epsilon_{r2}}} \text{ (Np/m)}$$
$$\beta_2 \approx \omega \sqrt{\mu_2 \epsilon_2} = \frac{\omega \sqrt{\epsilon_{r2}}}{c} \text{ (rad/m)}$$
$$\eta_2 \approx \frac{\eta_0}{\sqrt{\epsilon_{r2}}} \angle \approx 0$$

$\epsilon = \epsilon_r \epsilon_0$; $\epsilon_0 = 8.85 \times 10^{-12}$ F/m; $36\pi \epsilon_0 \approx 1 \times 10^{-9}$.

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

Multilayer chain *stack of slabs 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4*

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

Air-slabs-air shortcut *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT

Beam offset through N slabs *thickness t_i , refracted angle θ_i*

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

Each slab adds $t_i \tan \theta_i$. Final air-to-air exit returns to θ_1 but is laterally shifted by d .

Common values $\arctan$

$\sin \theta$	θ
0.2	11.54°
1/3	19.47°
1/2	30°
0.5774	35.26°
$1/\sqrt{2}$	45°

FRESNEL — BRIEF REFERENCE

Perpendicular pol. ($E \perp$ plane of inc.)

$$\Gamma_{\perp} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t}$$

Parallel pol. ($E \parallel$ plane of inc.)

$$\Gamma_{\parallel} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

$\tau_{\perp} = 1 + \Gamma_{\perp}$, $\tau_{\parallel} = (1 + \Gamma_{\parallel}) \cos \theta_i / \cos \theta_t$

Brewster ($\parallel$ only): $\tan \theta_B = \sqrt{\epsilon_{r2}/\epsilon_{r1}} \Rightarrow$ no reflection.

TRIG SHORTCUTS

Expr	=	Use
$\sin^{-1}(0.2)$	0.2014 rad	θ_2 in HW3 P3
$\sqrt{1 - (f_c/f)^2}$	u_g/c	waveguide ratio
$\sqrt{1/a^2 + 1/b^2}$	$\sqrt{(b^2 + a^2)/(ab)^2}$	TE ₁₁ cutoff
$\pi/4 = 45^\circ$, $\pi/6 = 30^\circ$, $\pi/3 = 60^\circ$, $\pi/2 = 90^\circ$		

RECTANGULAR WAVEGUIDE — MODES

TE mode $E_z = 0, H_z \neq 0$

TM mode $H_z = 0, E_z \neq 0$

Dimensions $a \times b$ with $a > b$ along $\hat{x}, \hat{y}$; wave in $\hat{z}$.

E_x form (TE) *read m, n from arguments*

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on $x = m\pi/a$; coef. on $y = n\pi/b$.

TE mode allowed if at least one of $m, n \geq 1$. TM needs both $m, n \geq 1$.

CUTOFF FREQUENCY

f_{mn} m, n integers ≥ 0 (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$
$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} \text{ (m/s, guide medium)}$$

Common cutoffs ($a > b$)

Mode	f_c
TE ₁₀	$u_{p0}/(2a)$ (dominant)
TE ₀₁	$u_{p0}/(2b)$
TE ₂₀	$u_{p0}/a = 2f_{10}$
TE ₁₁ , TM ₁₁	$\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$

Propagation requires $f > f_{mn}$.

ϵ_r FROM DISPERSION

Given ω, β , mode (m, n)

$$\epsilon_r = \frac{c^2}{\omega^2} \left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right]$$

Sanity: β in rad/m, a, b in m, ω in rad/s.

WAVE IMPEDANCE & H_y

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$
$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

$Z_{TE} > \eta$ always (denominator < 1); $Z_{TM} < \eta$.

GROUP VELOCITY & TRAVEL TIME

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$
$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \Rightarrow u_p u_g = u_{p0}^2$$

$t = L/u_g$ (s) travel time through guide of length L
Higher modes have lower $u_g \Rightarrow$ they arrive later. Group delay spread = sum over excited modes.

BOUNDARY

η_1, η_2 — imp. (Ω)

Γ — refl. coeff (unitless)

$\tau = 1 + \Gamma$ — trans. coeff

$k_i = \omega \sqrt{\epsilon_{r,i}}/c$ (rad/m)

α_2, β_2 — if med 2 lossy

$z = 0$ — boundary plane

$\eta_0 = 120\pi \approx 377 \Omega$

SNELL / OBLIQUE

θ_i, θ_t — angles from normal

$n_i = \sqrt{\epsilon_{r,i}}$ — index of refr.

t_i — slab thickness (m)

d — lateral disp. (m)

plane of inc. = plane of $\hat{k}$ +normal

$\perp$: $\mathbf{E} \perp$ this plane

$\parallel$: $\mathbf{E} \parallel$ this plane

WAVEGUIDE

a, b — wide, narrow dim. (m); $a > b$

m, n — mode indices (int.)

f_{mn} — cutoff (Hz)

β — prop. const (rad/m)

Z_{TE}/Z_{TM} — wave imp. (Ω)

E_x, H_y — field comps (V/m, A/m)

$\eta = \eta_0/\sqrt{\epsilon_r}$

VELOCITIES

$u_{p0} = c/\sqrt{\epsilon_r}$ — bulk phase vel.

$u_p = u_{p0}/\sqrt{1 - (f_c/f)^2}$

$u_g = u_{p0} \sqrt{1 - (f_c/f)^2}$

$u_p u_g = u_{p0}^2$

$t = L/u_g$ — travel time

$c = 3 \times 10^8$ m/s

POWER & UNITS

$\%P_r = 100|\Gamma|^2$

$\%P_t = 100|\tau|^2 \eta_1/\eta_2$

$\%P_r + \%P_t = 100$

σ (S/m), ϵ (F/m), μ (H/m)

$\epsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 = 4\pi \times 10^{-7}$ H/m

$1/(36\pi) \times 10^{-9} \approx \epsilon_0$

DIPOLE TYPE BY LENGTH

Check first <i>compute $\lambda = c/f$, then l/λ</i>		
l/λ	Type	Use
$< 1/50$	Hertzian (short)	Hertzian formula
$= 1/4$	Quarter-wave	Arbitrary formula
$= 1/2$	Half-wave	$D = 1.64$, simple
other	Arbitrary	Arbitrary formula

If $l/\lambda < 1/50$ stop and use Hertzian only — don't subst. small l into arbitrary formula.

HERTZIAN DIPOLE — $S(R, \theta)$

Power density <i>$l/\lambda < 1/50$; far field</i>
$S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32 \pi^2 R^2} \sin^2 \theta \text{ (W/m}^2\text{)}$
$k = 2\pi/\lambda \text{ rad/m; } I_0 \text{ peak current (A); } l \text{ length (m); } R \text{ obs. dist (m).}$
Max direction $\theta = \pi/2$ (broadside)
Total radiated power <i>integrate over $4\pi \text{ sr}$</i>
$P_{\text{rad}} = \frac{\eta_0 \pi}{3} \left(\frac{I_0 l}{\lambda} \right)^2 \text{ (W)}$
Directivity <i>Hertzian, exact</i>
$D_{\text{Hertz}} = 1.5 \text{ (i.e. 1.76 dB)}$

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at distance R <i>any l</i>
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[\frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$
Normalized pattern $F(\theta) = S(\theta)/S_{\text{max}}$, <i>dimensionless</i>
Half-wave $l = \lambda/2 \quad \pi l/\lambda = \pi/2$
$S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}, \quad D = 1.64, \quad R_{\text{rad}} = 73 \Omega$
Quarter-wave $l = \lambda/4 \quad \pi l/\lambda = \pi/4, \cos(\pi/4) = \sqrt{2}/2$
$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2, \quad S_0 = \frac{15 I_0^2}{\pi R^2}$
$S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2} \right)^2 \approx 0.0858 S_0$

BEAM PARAMETERS — HPBW, Ω_p, D

1. Normalize
$F(\theta) = S(\theta)/S_{\text{max}}$
2. Half-power beamwidth (HPBW) <i>solve $F(\theta_{1/2}) = 0.5$</i>
$\beta = 2\theta_{1/2} \text{ (symmetric beams)}$
example $F(\theta) = e^{-20\theta^2} \Rightarrow \theta_{1/2} = \sqrt{-\ln 0.5/20} = 0.186 \text{ rad} \Rightarrow \beta = 21.31^\circ$
3. Pattern solid angle <i>(sr)</i>
$\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta \, d\theta \, d\phi$
If F has no ϕ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta \, d\theta$
4. Directivity
$D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$
5. Gain $\xi = \text{radiation efficiency, } 0 < \xi \leq 1$
$G = \xi D \quad G_{\text{dB}} = 10 \log_{10} G$
<i>For lossless / ideal antennas: $G = D$.</i>

COMMON DIPOLE PARAMETERS

Type	D	D_{dB}	$R_{\text{rad}} \text{ (}\Omega\text{)}$
Isotropic	1	0	—
Hertzian	1.5	1.76	$80\pi^2 (l/\lambda)^2$
Half-wave	1.64	2.15	73
$l = \lambda$	2.41	3.82	199

EFFECTIVE AREA A_e

Definition m^2 , <i>antenna's receiving cross section</i>
$A_e = \frac{\lambda^2 D}{4\pi}$
Physical cross section <i>half-wave dipole, wire diameter d</i>
$A_p = l d = \frac{\lambda}{2} d$
Ratio $A_e \gg A_p$ <i>for thin wires (good!)</i>
Example: $f = 100 \text{ MHz}, d = 2 \text{ cm} \Rightarrow A_e/A_p \approx 39$.

POWER FLOW IDENTITIES

$P_{\text{rad}} = S_{\text{max}} \Omega_p R^2 = \frac{4\pi R^2 S_{\text{max}}}{D}$
$S_{\text{max}} = \frac{P_{\text{rad}} D}{4\pi R^2} \text{ (W/m}^2\text{)}$
<i>For isotropic radiator: $S = P_{\text{rad}}/(4\pi R^2)$.</i>

FRIIS TRANSMISSION FORMULA

Far-field link budget <i>P_t tx, P_r rx, dist R</i>
$P_r = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2 P_t$
G_t, G_r linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.
Solve for R
$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}}$
Solve for G_r given A_e
$G_r = \frac{4\pi A_{e,r}}{\lambda^2} \Rightarrow P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}$
<i>Use linear G, not dB. Convert first: $G_{\text{lin}} = 10^{G_{\text{dB}}/10}$.</i>
<i>Free-space path loss $L_{\text{fs}} = (4\pi R/\lambda)^2$ so $P_r = G_t G_r P_t / L_{\text{fs}}$.</i>

dB $\leftrightarrow$ LINEAR

Power gain <i>factor $\rightarrow$ dB</i>
$G_{\text{dB}} = 10 \log_{10} G_{\text{lin}} \quad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$
dB linear dB linear
0 1 10 10
3 2 13 20
6 4 20 100
7 5 30 1000
−3 0.5 −10 0.1
<i>For E or amplitude use $20 \log$ not $10 \log$.</i>

RADAR EQUATION (BONUS)

Monostatic radar <i>$tx = rx$, target cross section $\sigma_t \text{ (m}^2\text{)}$</i>
$P_r = \frac{P_t G^2 \lambda^2 \sigma_t}{(4\pi)^3 R^4}$
Doppler shift <i>closing velocity v_r</i>
$f_d = \frac{2v_r}{\lambda} = \frac{2v_r f}{c} \text{ (Hz)}$
<i>Factor of 2 from round-trip path; v_r is the radial component toward radar.</i>

LECTURE-11 BOUNDARY $\leftrightarrow$ T-LINE

EM	T-line	Match
$\tilde{E}_i$	$\tilde{V}_1^+$	forward wave
$\tilde{E}_r$	$\tilde{V}_1^-$	reverse wave
η_i	Z_{0i}	line impedance
Γ	Γ	same formula
k_i	β_i	wavenumber

Bouncing-wave analogy: every t-line formula in Ch. 2 transfers verbatim with the substitutions above.

ANTENNA INTEGRATION TIPS

<i>Many radiation integrals (e.g. Ω_p for $e^{-20\theta^2} \sin \theta$) have no closed form.</i>
<i>Use Wolfram / numerical integration. State this in the answer.</i>
<i>Common shortcut: if $F(\theta)$ is narrow ($\theta_{1/2} \ll 1$), approximate</i>
$\Omega_p \approx \theta_{1/2}^2 \pi / (4 \ln 2) \text{ (Gaussian beam)}$

DIPOLE

l — antenna length (m)
$\lambda = c/f$ — wavelength (m)
I_0 — peak current (A)
$\eta_0 = 120\pi \approx 377 \Omega$
$k = 2\pi/\lambda$ — wavenumber (rad/m)
R_{rad} — radiation res. (Ω)
R — obs. dist (m)

RADIATION

$S(R, \theta)$ — power density (W/m^2)
S_{max} — peak power density
$F(\theta) = S/S_{\text{max}}$ — normalized
Ω_p — pattern solid angle (sr)
β — HPBW (rad or deg)
D — directivity (unitless)
P_{rad} — total rad. power (W)

ANTENNA

$A_e = \lambda^2 D / (4\pi)$ — effective area
A_p — physical cross sec. (m^2)
ξ — rad. efficiency ($0 < \xi \leq 1$)
$G = \xi D$ — gain (unitless)
$G_{\text{dB}} = 10 \log G$
Half-wave: $D = 1.64, R_{\text{rad}} = 73 \Omega$
Hertzian: $D = 1.5$

FRIIS / LINK

P_t — tx power (W)
P_r — rx power (W)
G_t, G_r — linear gains
R — range (m)
$L_{\text{fs}} = (4\pi R/\lambda)^2$ — path loss
3 dB $\rightarrow$ 2; 10 dB $\rightarrow$ 10
$G_{\text{lin}} = 10^{G_{\text{dB}}/10}$

RADAR

σ_t — RCS (m^2)
f_d — Doppler shift (Hz)
v_r — radial vel. (m/s)
$P_r \propto 1/R^4$ (monostatic)
$f_d = 2v_r/\lambda$ (round trip)
$c = 3 \times 10^8 \text{ m/s}$
$d > 0$: target approaching